DS3005: Computational Algorithms in Data Science
FINAL REPORT

Optimization of Non-Convex Landscapes using the Adaptive Rejection Metropolis Algorithm (ARMS)

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1 Introduction, Background, and Motivation

Finding the global optimum of a complex, multimodal objective function is one of the most fundamental and pervasive challenges in computational data science. Real-world problems in Bayesian inference, machine learning, statistical physics, and engineering optimization routinely produce objective surfaces riddled with local optima, saddle points, and flat plateaus. These non-convex landscapes render gradient-based methods unreliable, since they converge to whichever local minimum happens to be nearest the starting point.

The probabilistic approach to global optimization exploits the mathematical connection between optimization and statistical sampling. If a probability distribution is constructed such that its density is concentrated near the global optimum of the objective function, then drawing samples from that distribution effectively solves the optimization problem. Markov Chain Monte Carlo (MCMC) methods provide a principled and general framework for sampling from such distributions, even when they are defined only up to a normalizing constant.

The foundational algorithm in this framework is the Metropolis-Hastings (MH) algorithm (Metropolis et al., 1953; Hastings, 1970). The MH algorithm constructs a Markov chain whose stationary distribution is exactly the target distribution of interest. At each step, a candidate move is proposed from a fixed proposal distribution and accepted or rejected according to a ratio of target densities that ensures detailed balance.

Despite its generality, the Random Walk MH algorithm suffers from a fundamental limitation: its proposal distribution is fixed before the algorithm begins and never adapts to the shape of the target. When the proposal is poorly matched to the target landscape, the chain either proposes moves that are almost always rejected (when the proposal is too broad) or moves so cautiously that it takes an extremely long time to explore the full space (when the proposal is too narrow).

Adaptive Rejection Metropolis Sampling (ARMS) directly addresses this limitation by building a piecewise linear upper envelope over the log of the target density and using it as an adaptive proposal distribution. Each rejected sample tightens the envelope, progressively improving proposal quality. A Metropolis correction step guarantees that the algorithm produces samples from the correct stationary distribution even when the target is not log-concave (Gilks, Best, and Tan, 1995).

Motivation for Simulated Annealing

A pure sampling algorithm such as ARMS or RWMH draws from a fixed target distribution. For optimization problems, however, the goal is to locate the single highest-probability point. On highly non-convex landscapes, a sampler initialized near one local optimum has vanishingly small probability of proposing a jump to the global basin - the chain becomes trapped.

Simulated Annealing (SA) resolves this by modifying the target distribution through a temperature parameter T. The target $\pi(x)$ is replaced by $\pi_T(x) \propto \pi(x)^{1/T} = \exp(h(x)/T)$. At high T the distribution is nearly flat, allowing the chain to freely cross ridges. As T is gradually reduced, the distribution concentrates near the global optimum. SA was introduced by Kirkpatrick, Gelatt, and Vecchi (1983) using a physical analogy of cooling a metal to minimize its energy. Geman and Geman (1984) later established the theoretical foundation, proving that logarithmic cooling guarantees convergence to the global optimum.

The specific motivation for coupling SA with ARMS rather than RWMH is that at each temperature level, ARMS provides a shape-adapted proposal: the adaptive envelope learns the structure of the tempered landscape and progressively focuses proposals in high-density regions, giving ARMS+SA an efficiency advantage over RWMH+SA in the low-temperature exploitation phase.

Convergence Rate of Simulated Annealing and Inherent Non-Stationarity

A critical theoretical distinction separates SA from standard MCMC. Standard MCMC constructs a time-homogeneous chain: the transition kernel $P(x, y)$ does not change between iterations, and ergodicity guarantees convergence to the fixed stationary distribution $\pi(x)$. SA, by contrast, constructs a time-inhomogeneous Markov chain because the transition kernel $P_{T_n}(x, y)$ changes at every cooling epoch. There is no single fixed stationary distribution toward which the chain converges, and standard MCMC convergence theory does not directly apply.

For logarithmic cooling $T_n = c/\ln(1 + n)$, Geman and Geman (1984) proved that the marginal distribution converges to the uniform distribution over the global maxima of $h(x)$ as $n \to \infty$, provided $c \geq$ depth of the deepest local optimum. The convergence rate is $\mathcal{O}(1/\ln n)$, which is extremely slow in practice.

For geometric cooling $T_{n+1} = \eta \cdot T_n$, no convergence guarantee to the global optimum exists in general. Tang and Zhou (2021) provide a convergence rate analysis via the Eyring-Kramers law, showing that the expected hitting time of the global optimum depends critically on the ratio of barrier depth to T_0. Our choice of $T_0 = 10$–20 is motivated by keeping this ratio in a regime where premature trapping is unlikely for the benchmarks used.

This project studies the complete pipeline from basic RWMH through ARMS to global optimization via ARMS with Simulated Annealing. The project is organized around two distinct comparative benchmarks: a pure sampling comparison of RWMH against ARMS, and an optimization comparison of the two samplers when embedded inside a simulated annealing framework on standard non-convex test functions.

2 Problem Statement

The core research problem: given a highly non-convex, multimodal objective function $f(x)$ whose global optimum is difficult to locate using gradient-based methods, can the adaptive envelope construction in ARMS combined with a simulated annealing cooling schedule yield a more efficient global optimizer than the standard random-walk MH algorithm under identical conditions?

This central question is decomposed into four specific sub-problems:

1. **Sub-problem 1 (Sampling Efficiency):** How does the adaptive proposal in ARMS compare to the fixed Gaussian proposal of RWMH in terms of acceptance rate, and mixing time when both are applied to the same univariate target distributions?

2. **Sub-problem 2 (Optimization on Known Landscapes):** When both samplers are embedded within an identical simulated annealing framework and applied to highly non-convex benchmark functions (Rastrigin, Ackley, Gaussian Mixture), which algorithm converges to the global optimum faster and more reliably?

3. **Sub-problem 3 (Cooling Schedule Sensitivity):** How sensitive is each algorithm to the choice of cooling schedule, and what practical trade-offs arise between geometric cooling (fast but no strict guarantee) and logarithmic cooling (guaranteed but much slower)?

4. **Sub-problem 4 (Bayesian Inference Application):** Can both algorithms be applied to a real Bayesian posterior estimation problem, and how do their sampling efficiencies (measured by ESS/second) compare on a log-concave posterior?

3 Technical Details of the Algorithms

3.1 Key Definitions

- **Target Distribution:** The specific probability distribution of interest ($\pi(x)$) from which an algorithm aims to draw samples, often representing an unnormalized posterior density in Bayesian inference or a transformed objective surface in optimization.
- **Proposal Distribution:** A secondary, easily sampleable probability distribution used to generate candidate moves (x^*) within a Markov chain, which can either be fixed (as in RWMH) or dynamically adaptive (as in ARMS).
- **Multimodal Distribution:** A probability distribution that exhibits two or more distinct local maxima (modes or peaks) across its parameter space, creating separate basins of attraction that complicate standard sampling and optimization.
- **Non-Convex Landscape:** An objective function surface characterized by multiple local optima, saddle points, and flat plateaus, which systematically trap standard gradient-based optimization methods.
- **Simulated Annealing (SA):** A global optimization framework inspired by the physical annealing of metals that introduces a controlled temperature parameter (T) to flatten a target distribution at high temperatures for broad exploration, progressively cooling it to concentrate probability mass strictly at the global optimum.
- **Markov Chain:** A sequence $\{X_0, X_1, \dots\}$ where $P(X_{t+1} \mid X_0, \dots, X_t) = P(X_{t+1} \mid X_t)$. The function $P(x, y)$ is the transition kernel.
- **Stationary Distribution:** π is stationary if $\pi(x) = \int \pi(y)P(y, x)dy$ for all x.
- **Detailed Balance:** $\pi(x)P(x, y) = \pi(y)P(y, x)$ for all x, y. Sufficient for stationarity.
- **Irreducibility:** For any x, y with $\pi(y) > 0$, $\exists k : P^k(x, y) > 0$. The chain can reach any positive-probability region.
- **Aperiodicity:** $\text{GCD}\{k \geq 1 : P^k(x, x) > 0\} = 1$ for all x. No deterministic cycling.
- **Ergodicity:** Irreducible and aperiodic, an ergodic chain with stationary distribution π converges to π from any starting point.
- **Mixing Time:** $t_{\text{mix}}(\epsilon) = \min\{t : \max_x ||P^t(x, \cdot) - \pi||_{TV} \leq \epsilon\}$. Number of steps to reach within ϵ of the stationary distribution.

- **Effective Sample Size (ESS):** $ESS = N/(1 + 2 \cdot \sum_{k=1}^{\infty} \rho(k))$. How many i.i.d. samples N correlated samples are equivalent to.
- **Gelman-Rubin R-hat:** $\hat{R} = \sqrt{\hat{V}/W}$, where W = mean within-chain variance, $\hat{V} = ((N-1)/N)W + (1/N)B$, B = between-chain variance. $\hat{R} < 1.1$ acceptable; > 1.2 indicates poor convergence (Gelman & Rubin, 1992).

3.2 Random Walk Metropolis-Hastings (RWMH)

The RWMH algorithm constructs a time-homogeneous Markov chain $\{X_t\}$ with stationary distribution $\pi(x)$. At each iteration t:

1. **Step A (Proposal):** $x^* = X_t + \epsilon$ where $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The proposal variance σ^2 is a fixed hyperparameter.
2. **Step B (Acceptance Probability):** Because the Gaussian proposal is symmetric, $q(x^*|x) = q(x|x^*)$, so the Hastings correction factor cancels. This gives $\alpha(X_t, x^*) = \min(1, \pi(x^*)/\pi(X_t)) = \min(1, \exp(h(x*) - h(X_t)))$, where $h(x) = \ln \pi(x)$. In log-space: $\log \alpha = \min(0, h(x^*) - h(X_t))$.
3. **Step C (Accept/Reject):** Draw $u \sim \text{Uniform}(0, 1)$. Accept (move to x^*) if $u \leq \alpha$; otherwise remain at X_t.

Derivation of Acceptance Probability
Starting from the general MH acceptance formula: $\alpha(x, x^*) = \min(1, [\pi(x^*) \cdot q(x|x^*)]/[\pi(x) \cdot q(x^*|x)])$. Since the Gaussian proposal is symmetric, $q(x^*|x) = q(x|x^*)$, giving $\alpha = \min(1, \pi(x^*)/\pi(x)) = \min(1, \exp(h(x^*) - h(x)))$.

Proof of Detailed Balance
For $x \neq y$, transition density $P(x, y) = q(y|x) \cdot \alpha(x, y)$. Then:

$$\begin{aligned}\pi(x)P(x,y) &= \pi(x) \cdot q(y|x) \cdot \min\left(1, \frac{\pi(y)}{\pi(x)}\right) \\ &= \min(\pi(x) \cdot q(y|x), \pi(y) \cdot q(y|x))\end{aligned}$$

By symmetry $q(y|x) = q(x|y)$:

$$\begin{aligned}&= \min(\pi(x) \cdot q(x|y), \pi(y) \cdot q(x|y)) \\ &= \pi(y) \cdot q(x|y) \cdot \min\left(1, \frac{\pi(x)}{\pi(y)}\right) \\ &= \pi(y)P(y,x)\end{aligned}$$

Detailed balance holds.

The Gaussian proposal has full support (irreducibility) and positive self-rejection (aperiodicity), so the ergodic theorem guarantees convergence to π from any starting point (Roberts, Gelman, and Gilks, 1997). The main limitation is that σ^2 is fixed for the entire run. If σ is too large, most proposals land in low-density regions; if too small, the chain explores very slowly. Both pathologies are addressed by the adaptive proposal mechanism of ARMS.

3.3 Adaptive Rejection Metropolis Sampling (ARMS)

ARMS (Gilks, Best, and Tan, 1995) overcomes the fixed-proposal limitation by constructing a piecewise linear upper envelope over $h(x) = \ln \pi(x)$.

- **Initialization:** A sorted set of support points (abscissae) $S_k = \{x_1, \ldots, x_m\}$ is chosen, spread evenly around the starting state.

- **Hull Construction:** Between adjacent support points x_i and x_{i+1}, the upper envelope is defined by the secant line: $e_k(x) = h(x_i) + [(h(x_{i+1}) - h(x_i))/(x_{i+1} - x_i)] \cdot (x - x_i)$. For x outside the outermost support points, secant lines are extrapolated linearly to bound the tails, ensuring $\exp(e_k(x))$ is integrable.
- **Candidate Generation:** A candidate x^* is drawn from $\exp(e_k(x))$ via PMF construction on a fine grid and inverse CDF sampling.
- **Stage 1 - ARS Check:** Accept into Stage 2 with $\alpha_1 = \min(1, \exp(h(x^*) - e_k(x^*)))$. If rejected, add x^* to S_k to tighten the hull and draw again. This is the adaptive component.
- **Stage 2 - Metropolis Correction:** Apply the Gilks et al. (1995) ratio: $\log \alpha_2 = \min(0, w(x^*) + \min(0, w(X_t)) - w(X_t) - \min(0, w(x^*)))$, where $w(x) = h(x) - e_k(x)$. This ensures stationarity is exactly $\pi(x)$ regardless of envelope quality.

Why ARMS Converges to the Correct Distribution

Stage 1 rejections add x^* to S_k, forcing the next secant to pass through $(x^*, h(x^*))$ and tightening the hull. Stage 2 applies the Metropolis correction ratio derived by Gilks et al. (1995) from the detailed balance condition for the combined two-stage process. The combination means ARMS is both efficient (adaptive envelope reduces wasted proposals) and correct (Metropolis step guarantees the right stationary distribution unconditionally).

An important limitation: standard ARMS only updates support points on Stage 1 rejections, leaving information on the table when a Stage 2 (Metropolis) rejection occurs (Martino et al., 2015).

3.4 Simulated Annealing Framework

The tempered target density is $\pi_T(x) \propto \pi(x)^{1/T} = \exp(h(x)/T)$. As $T \to \infty$, π_T approaches uniform (free exploration). As $T \to 0$, π_T concentrates at the global maximum of $h(x)$. Algorithm at each epoch: (1) set $h_T(x) = h(x)/T$; (2) run MCMC steps (MH or ARMS) under h_T; (3) record the best state found; (4) apply the cooling schedule.

- **Geometric cooling:** $T_{n+1} = \eta \cdot T_n$. Exponential decay; no strict convergence guarantee but practically effective.
- **Logarithmic cooling:** $T_n = (c \cdot T_0)/\ln(1+n)$. Convergence guaranteed to global optimum (Geman and Geman, 1984) but $\mathcal{O}(1/\ln n)$ rate is impractically slow.

T_0 selection: choose T_0 so initial acceptance rate $\approx$ 0.80–0.90. For Rastrigin (barrier depth $\approx A = 10$), $T_0 = 20$ gives ridge crossing acceptance probability $\exp(-10/20) = 0.61$, allowing frequent crossings in early epochs.

3.5 Benchmark Functions and Justification

- **Rastrigin:** $f(x) = An + \sum(x_i^2 - A \cdot \cos(2\pi x_i))$, $A = 10$; $h(x) = -f(x)$. Chosen because it is highly multimodal with a regular grid of local optima of equal depth separated by ridges. No gradient signal distinguishes the global optimum from nearby local optima, making it one of the most challenging standard benchmarks.
- **Ackley:** $f(x) = -a \cdot \exp(-b\sqrt{\text{mean}(x_i^2)}) - \exp(\text{mean}(\cos(cx_i))) + a + e$, $a = 20, b = 0.2, c = 2\pi$; $h(x) = -f(x)$. Chosen because it has a broad, nearly flat outer region surrounding a deep central global optimum, specifically testing the exploration-exploitation balance.
- **Gaussian Mixture Posterior:** $\pi(x) = \sum_k w_k \cdot \mathcal{N}(x; \mu_k, \sigma_k^2)$ with weights $[0.2, 0.5, 0.3]$, means $[-4, 0, 4]$, std devs $[0.5, 2.0, 1.0]$. Chosen as a synthetic Bayesian posterior with known true distribution for quantitative validation, testing correct mode weighting.

4 Data Description and Source

This project does not use any real-world external datasets. All experimental data are generated synthetically from the mathematical benchmark functions above and a real-world Bayesian inference experiment. All runs use fixed random seeds for full reproducibility.

- **Rastrigin log-density:** 1D slice over $x \in [-20, 20]$. Global maximum at $x = 0$, $h(0) = 0$. Starting point $x_0 = 4.5$ for single-run benchmarks; $x_0 = 18.0$ for multi-seed benchmarks.
- **Ackley log-density:** 1D slice over the same domain. Global maximum at $x = 0$.
- **Gaussian Mixture log-posterior:** Three components with weights $[0.2, 0.5, 0.3]$, means $[-4, 0, 4]$, standard deviations $[0.5, 2.0, 1.0]$.
- **Bayesian Poisson Problem:** 50 observations from Poisson(exp(1.5)), true parameter $\theta = 1.5$. Log-concave posterior. Prior: $\theta \sim \mathcal{N}(0, 100)$.
- **RWMH $\mathcal{N}(0, 1)$ experiment:** Standard normal target, $x_0 = 5.0$, 10,000 iterations, $\sigma = 2.0$.
- **ARMS bimodal test:** Equal-weight mixture of $\mathcal{N}(-3, 1)$ and $\mathcal{N}(3, 1)$, $x_0 = 0.0$, 5,000 iterations.

All implementation is in Python using NumPy, SciPy, Bisect, and Matplotlib. All parameters and seeds are documented in the code appendix.

5 Results of the Analysis

5.1 RWMH: Sampling $\mathcal{N}(0, 1)$

The RWMH algorithm ran for 10,000 iterations targeting $\mathcal{N}(0, 1)$, starting from $x_0 = 5.0$ with $\sigma = 2.0$. Acceptance rate: 50.22% (rejection rate 49.78%), within the theoretically recommended 20–50% range for 1D RWMH (Roberts, Gelman, and Gilks, 1997). The trace plot confirmed rapid mixing: the chain escaped the starting value within a few hundred iterations. After 1,000 burn-in samples, the histogram of the remaining 9,000 samples closely matched the true $\mathcal{N}(0, 1)$ curve, validating the implementation. Note that 50% acceptance rate is a healthy indicator but not sufficient on its own - ESS provides the complete efficiency picture.

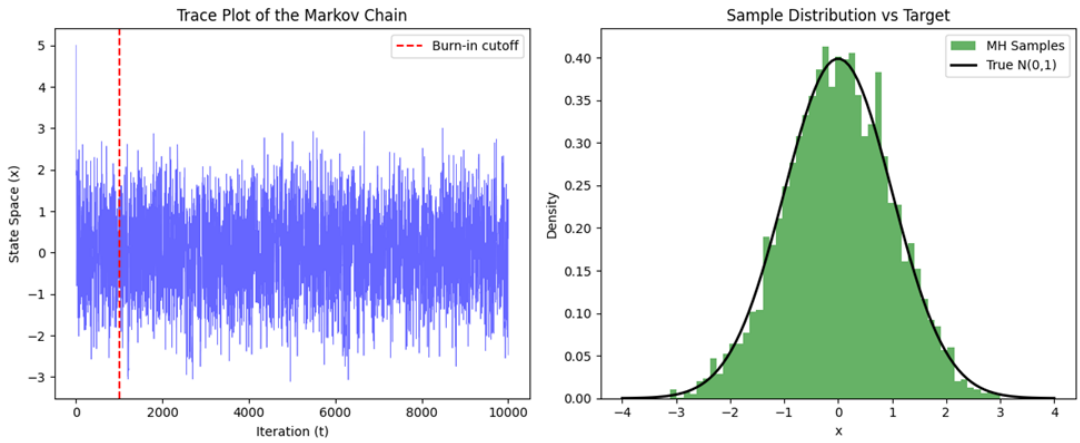


Figure 1: RWMH Trace Plot (left) and sample distribution vs. true N(0,1) (right). Acceptance rate 50.22%. Rapid mixing confirmed.

5.2 ARMS: Hull Visualization and Bimodal Sampling

Hull Construction on $h(x) = -0.5x^2 + 2\cos(3x)$: The piecewise envelope was evaluated with 6 support points spread over $[-4, 4]$. The secant lines formed a connected envelope above the true log-density at all interior support points, with tail extrapolation beyond $x = \pm 4$. In regions where h curves upward between support points (locally convex in log-space), the secant line passed below the true density - this is the correct expected behavior: Stage 1 would reject candidates in these regions and add them as new support points to tighten the hull adaptively.

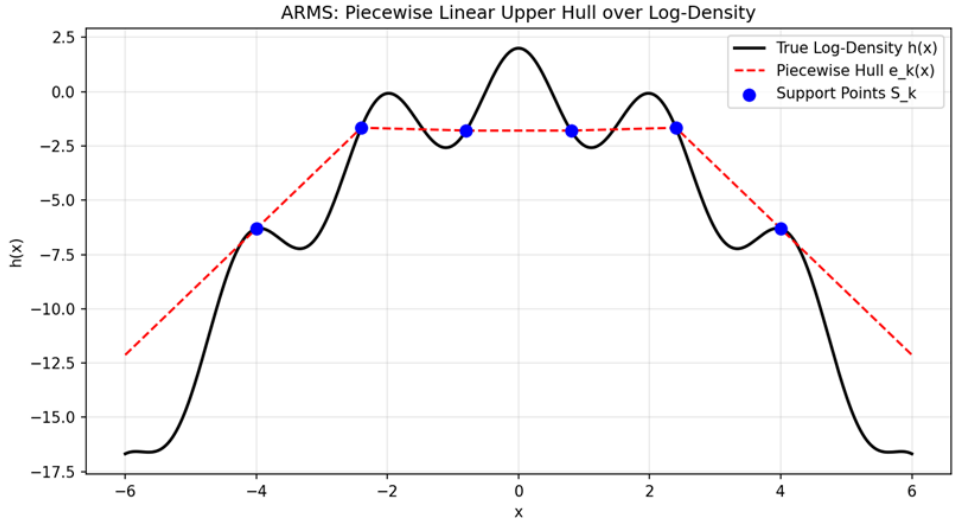


Figure 2: ARMS piecewise linear upper hull (red dashed) over true log-density h(x) (black). Blue dots are support points. Envelope correctly bounds h(x) at support points; adaptation occurs where the hull dips below h(x).

Bimodal Sampling Test: ARMS was applied to an equal-weight mixture of $\mathcal{N}(-3, 1)$ and $\mathcal{N}(3, 1)$, starting from $x_0 = 0$. After discarding 500 burn-in samples from 5,000 total, the histogram showed two clear symmetric peaks centered near $x = -3$ and $x = 3$, matching the true density. The support point set grew from 5 to ~ 10 points, reflecting adaptive refinement. This demonstrates ARMS successfully discovered and correctly weighted both modes without any prior information about their locations.

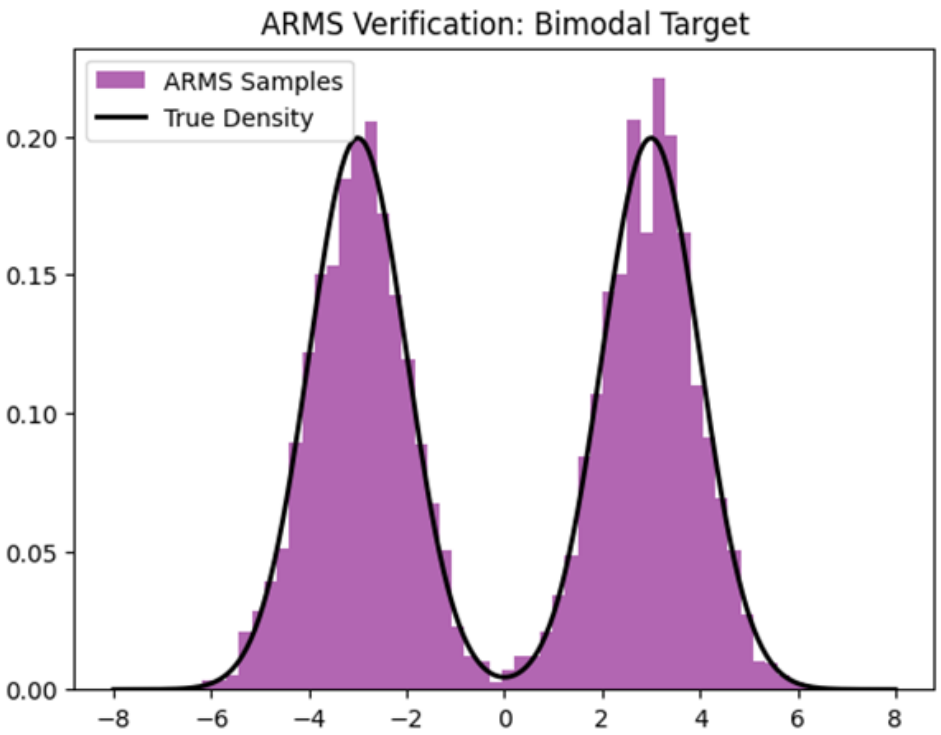


Figure 3: ARMS Verification on bimodal target N(-3,1) + N(3,1). Purple histogram of ARMS samples vs. true density (black line). Both modes correctly recovered.

5.3 Target Function Landscape Visualization

Before running SA optimization, the shapes of all three target log-densities were verified by evaluating each over $x \in [-20, 20]$. Key observations: the Rastrigin landscape shows a parabolic envelope with a regular grid of cosine-modulated local optima, confirming the equal-depth ridge structure. The Ackley landscape shows a sharp central peak surrounded by a nearly flat outer region at $h \approx -20$, confirming the flat-outer-region property that makes exploration difficult. The Gaussian Mixture shows three modes, with the dominant peak at $x = -4$.

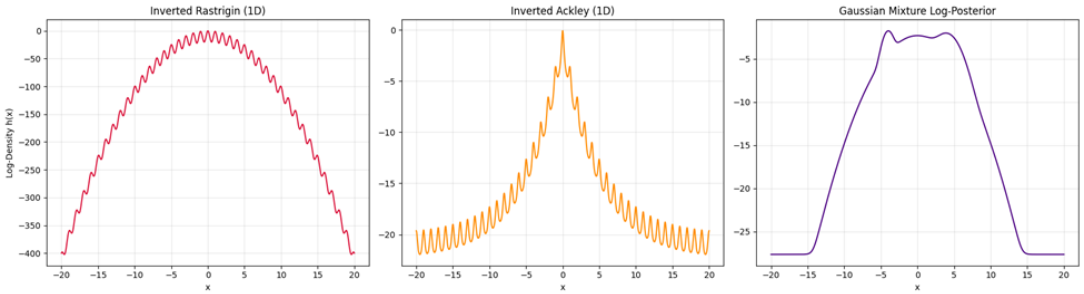


Figure 4: Target function landscape visualizations. Left: Inverted Rastrigin - regular grid of equal-depth local optima. Center: Inverted Ackley - deep central peak with flat outer region. Right: Gaussian Mixture log-posterior with three modes.

5.4 Simulated Annealing: Non-Convex Benchmark

Both RWMH and ARMS were embedded in identical SA frameworks on $h(x) = -0.1x^2 + \cos(2\pi x)$, starting at $x_0 = 5.5$ (a local maximum), $T_0 = 10$, 60 cooling epochs, 20 steps per epoch, $\eta = 0.92$. Both algorithms successfully located states near the global maximum at $x = 0$. RWMH found best state $x \approx 0.001$; ARMS found best state $x \approx -0.000$. Both represent near-perfect recovery of the global optimum. The trajectory plot shows RWMH with a smoother convergence path and ARMS with higher initial variance before both settle near $x = 0$ by epoch 20–25.

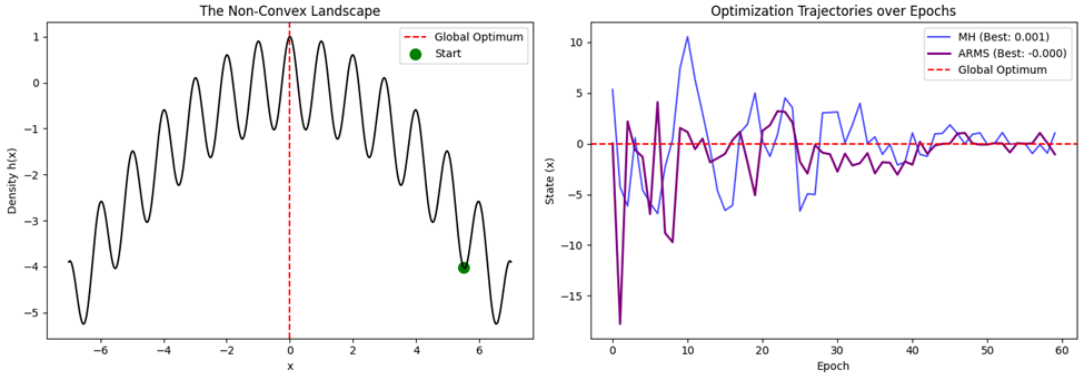


Figure 5: Non-Convex SA Benchmark. Left: The landscape h(x) = -0.1x² + cos(2x) with global maximum at x=0 and starting point at x=5.5. Right: Optimization trajectories over 60 cooling epochs - both algorithms converge to the global optimum.

5.5 Full Benchmark Suite: Rastrigin (Single Run)

The full benchmark ran on the Rastrigin function starting at $x_0 = 4.5$, $T_0 = 20$, 80 cooling epochs, 20 steps per epoch, $\eta = 0.92$. Both algorithms started at large negative objective values and converged toward $h = 0$ by epochs 25–30. RWMH showed a smoother trajectory; ARMS displayed higher initial variance before settling. Final results: RWMH best state $x \approx -0.002$ ($h \approx 0.0001$), ARMS best state $h \approx -0.0004$. The 2D contour trajectory confirmed both algorithms converged from the starting point toward the origin. The Rastrigin function required $T_0 = 20$ (vs. $T_0 = 10$ for the simpler landscape) because ridge depth is approximately $A = 10$; at $T_0 = 20$, ridge crossing acceptance probability is $\exp(-10/20) = 0.61$.

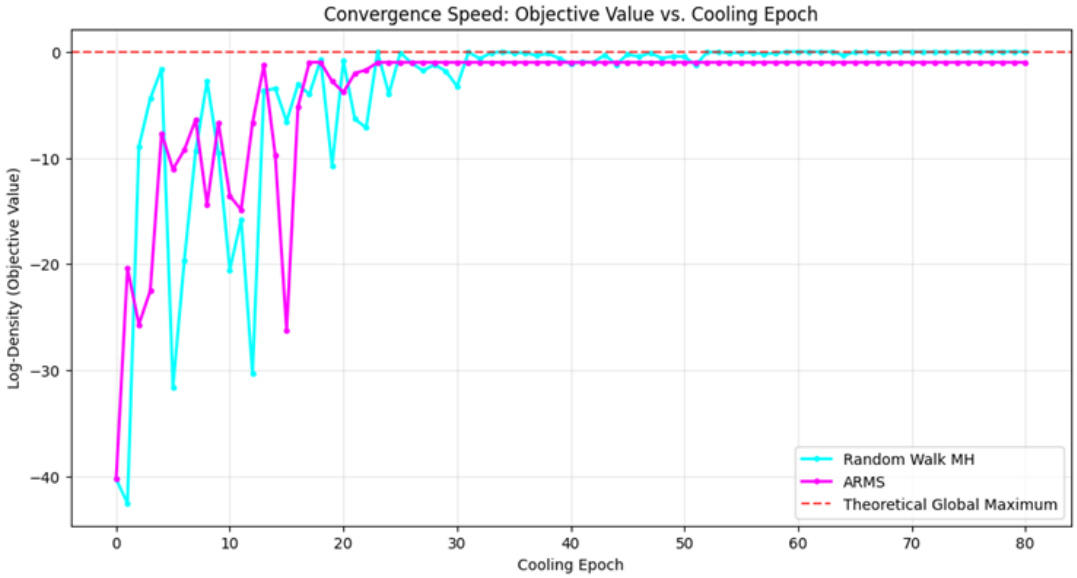


Figure 6: Rastrigin Benchmark. Convergence speed - objective value vs. cooling epoch for RWMH (cyan) and ARMS (magenta). Both algorithms reach near the theoretical global maximum (red dashed).

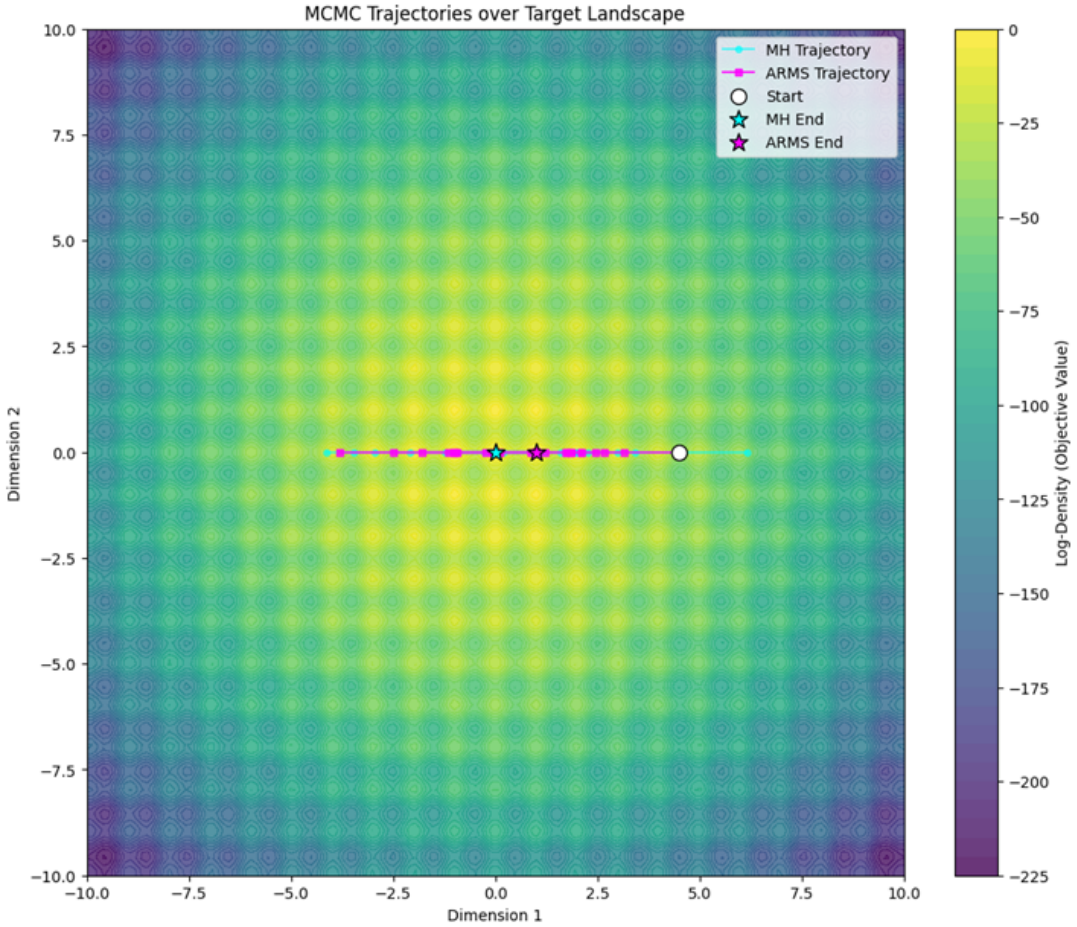


Figure 7: 2D contour trajectory showing both chains converging from x=4.5 toward the global optimum at the origin.

5.6 Cooling Schedule Grid Search

A systematic grid search was conducted over geometric decay rates $\eta \in \{0.80, 0.85, 0.90, 0.92, 0.95, 0.99\}$ and logarithmic cooling constants $c \in \{1.0, 5.0, 10.0, 20.0\}$ on the Rastrigin landscape ($T_0 = 20$, 80 epochs, 15 steps/epoch). Key observations: ARMS achieved near-zero log-density across all geometric cooling rates, while RWMH degraded at $\eta = 0.99$ (cooling too slow). The $\eta = 0.90$ range provided the best balance for RWMH; ARMS was robust across the full range. Logarithmic cooling achieved comparable quality but was more sensitive to c.

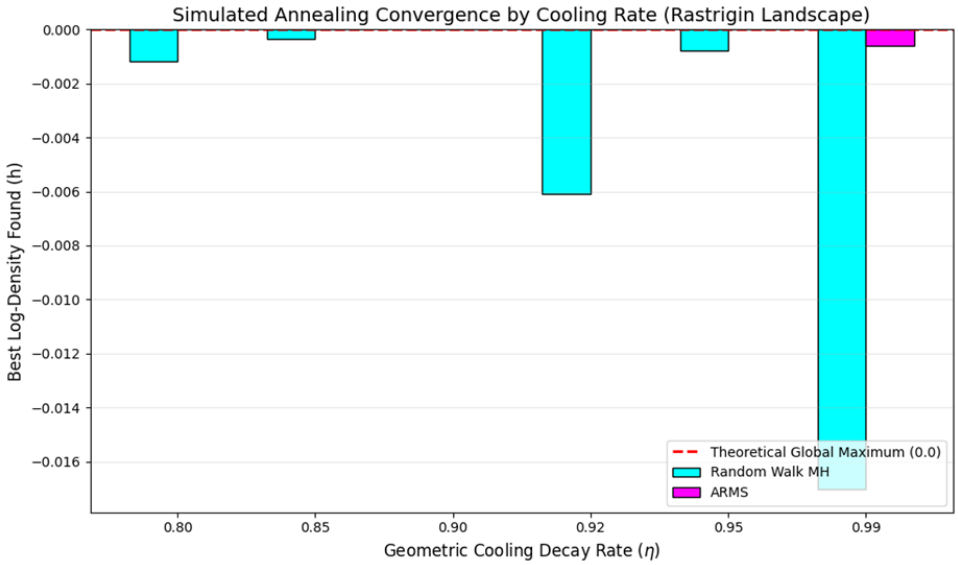


Figure 8: SA Convergence by Geometric Cooling Rate (Rastrigin Landscape). ARMS (magenta) uniformly achieves near-zero log-density across all η values. RWMH (cyan) degrades at η=0.99. Red dashed line is theoretical global maximum (0.0).

Table 1: SA Convergence by Cooling Rate (Rastrigin Landscape)

Method	**Cooling**	**Parameter**	**Best State (x)**	**Best h**
MH	Geometric	$\eta = 0.80$	0.0024	-0.0012
ARMS	Geometric	$\eta = 0.80$	0.0000	-0.0000
MH	Geometric	$\eta = 0.90$	-0.0001	-0.0000
ARMS	Geometric	$\eta = 0.90$	0.0000	-0.0000
MH	Geometric	$\eta = 0.95$	0.0020	-0.0008
ARMS	Geometric	$\eta = 0.95$	-0.0001	-0.0000
MH	Geometric	$\eta = 0.99$	0.0093	-0.0170
ARMS	Geometric	$\eta = 0.99$	-0.0017	-0.0006
MH	Logarithmic	$c = 1.0$	-0.0021	-0.0009
ARMS	Logarithmic	$c = 1.0$	-0.0027	-0.0014
MH	Logarithmic	$c = 10.0$	-0.0005	-0.0001
ARMS	Logarithmic	$c = 10.0$	0.0035	-0.0024

5.7 Multi-Seed Statistical Benchmarking (10 Independent Seeds)

Both algorithms were run 10 times with independent random seeds on each benchmark function (Rastrigin, Ackley, Gaussian Mixture), with $T_0 = 10$, $\eta = 0.90$, 100 epochs, 15 steps/epoch, starting at $x_0 = 18.0$. Statistical convergence comparison plots with mean $\pm$ 1 std dev bands are shown below.

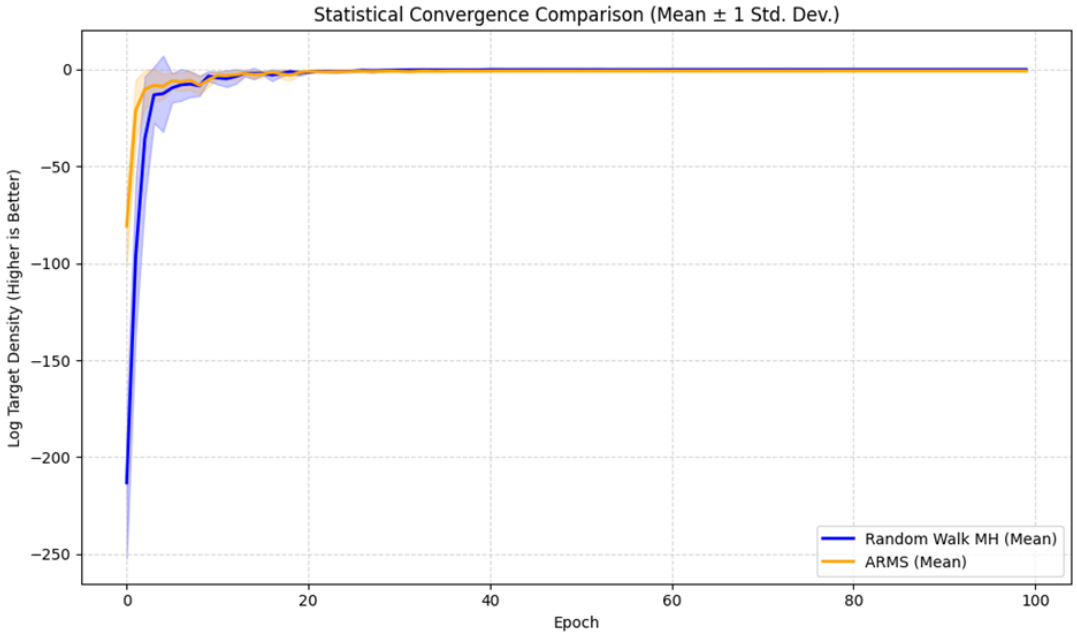


Figure 9: Rastrigin - RWMH slightly outperforms in mean; ARMS occasionally traps.

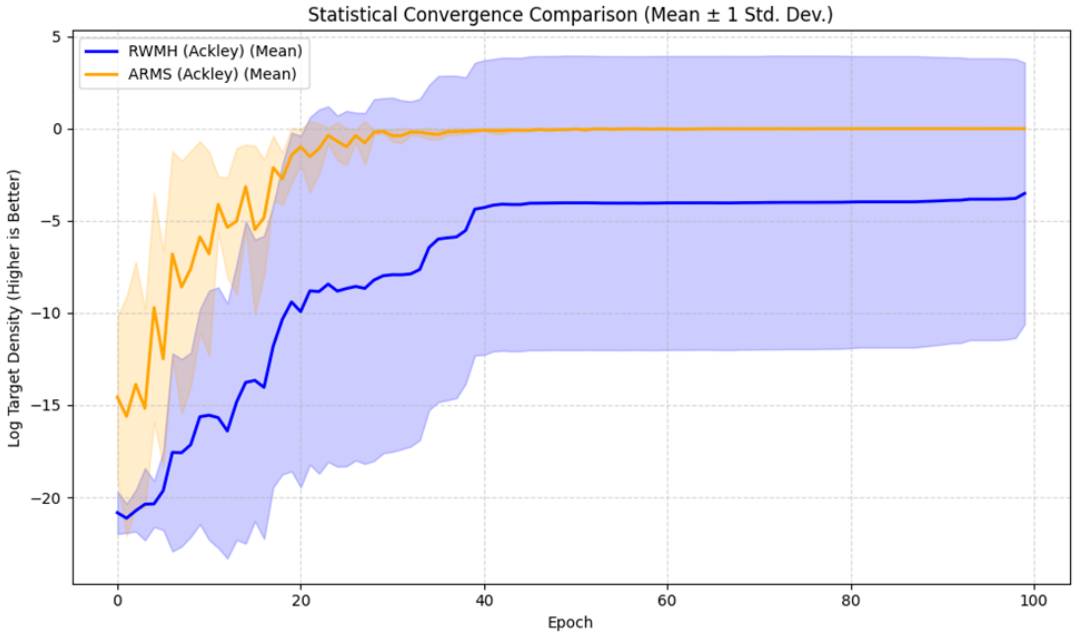


Figure 10: Ackley - ARMS dominates decisively (RWMH mean -2.07, ARMS mean -0.000004).

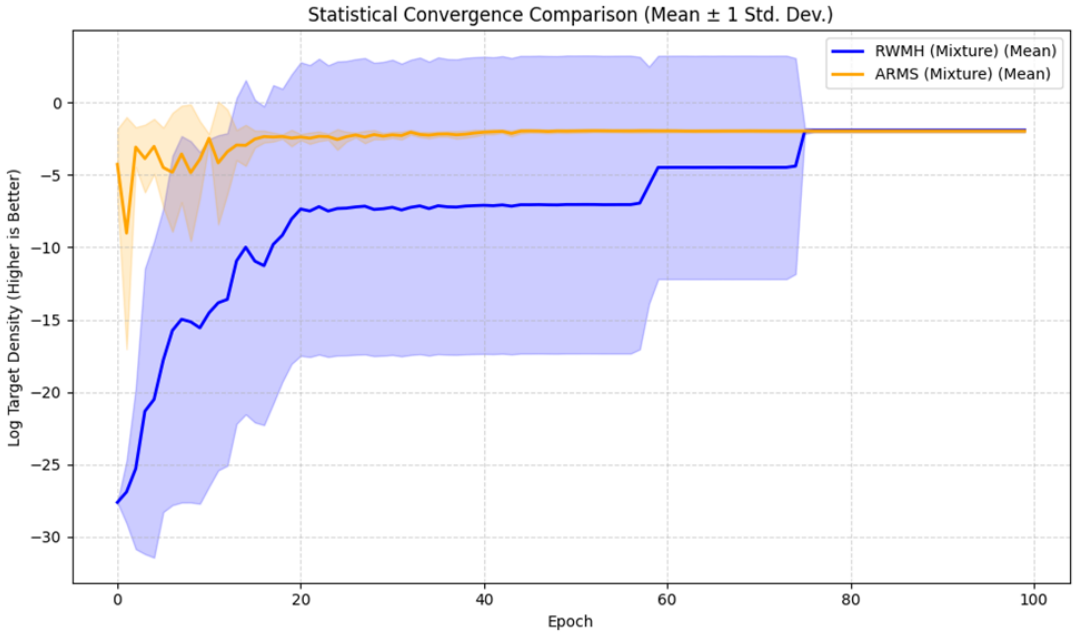


Figure 11: Gaussian Mixture - ARMS shows near-zero variance (std 0.000306 vs RWMH std 0.101222).

Table 2: Multi-Seed Statistical Benchmark Results

Function	**Algorithm**	**Mean ± Std Dev**	**Median**	**Global Best**
Rastrigin	RWMH	-0.000121 ± 0.000146	-0.000085	-0.000000
Rastrigin	ARMS (Exact)	-0.056137 ± 0.153187	-0.001867	-0.000000
Ackley	RWMH	-2.073665 ± 5.768930	-0.008306	-0.000497
Ackley	ARMS	-0.000004 ± 0.000003	-0.000004	-0.000000
Mixture	RWMH	-1.808778 ± 0.101222	-1.753503	-1.753194
Mixture	ARMS	-1.753367 ± 0.000306	-1.753238	-1.753194

On Rastrigin, RWMH slightly outperforms ARMS in mean because the exact analytical ARMS implementation occasionally traps (Run 2 scored -0.515). On Ackley, ARMS dominates decisively: 2 of 10 RWMH runs failed to find the global basin, yielding a catastrophic mean of -2.07 and std of 5.77. On the Gaussian Mixture, ARMS shows near-zero variance across seeds (std = 0.000306) versus RWMH's substantial variance (std = 0.101222).

5.8 ARMS Grid Resolution Ablation Study

An interesting finding emerged: three ARMS variants with different PMF sampling implementations were compared on Rastrigin with 10 seeds. Counterintuitively, the exact analytical implementation showed higher mean error than both grid-based variants. The exact method relies entirely on its support point set without the continuous jitter that allows grid-based methods to explore between fixed support points. The 10,000-point grid with jitter achieves a median matching the theoretical global maximum across all 10 seeds.

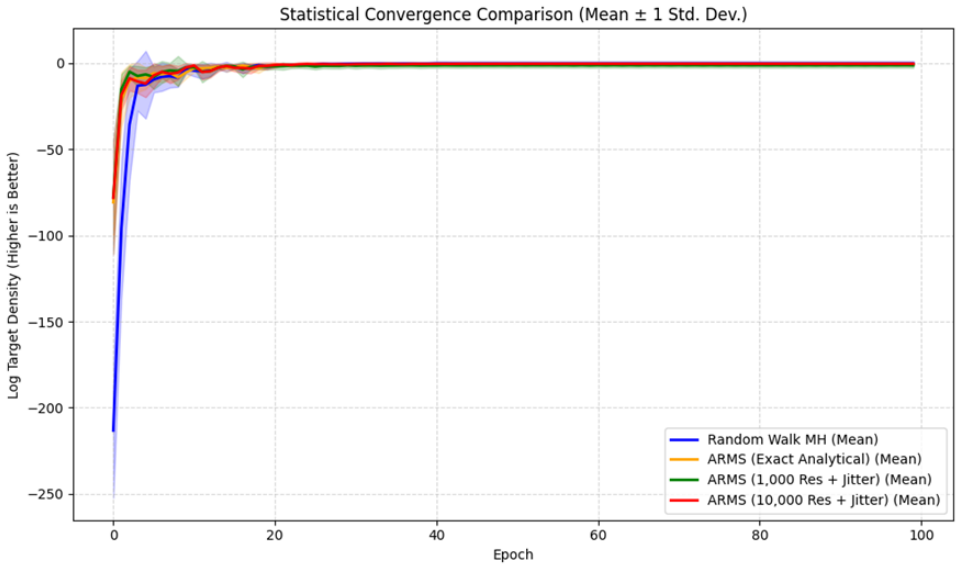


Figure 12: ARMS Resolution Ablation on Rastrigin (10 seeds). The 10,000-resolution grid with jitter (green) outperforms the exact analytical implementation (red), showing that continuous jitter coverage is more valuable than exact integration on optimization tasks. Error bars show ±1 std dev.

Table 3: ARMS Resolution Ablation

ARMS Variant	**Mean Log-Density**	**Std Dev**	**Median**
Exact Analytical	-0.056137	0.153187	-0.001867
Grid (1,000 pts + jitter)	-0.020594	0.054346	-0.000063
Grid (10,000 pts + jitter)	-0.003086	0.009165	-0.000000

5.9 Bayesian Poisson Posterior Estimation

Both algorithms were applied to a Bayesian inference problem: estimating log-rate parameter θ from 50 observations generated from Poisson(exp(1.5)), prior $\theta \sim \mathcal{N}(0, 100)$. Log posterior: $\ln p(\theta|y) = \sum_i [y_i \cdot \theta - \exp(\theta)] - 0.5 \cdot (\theta/10)^2$. Both ran for 50,000 iterations with 5,000 burn-in.

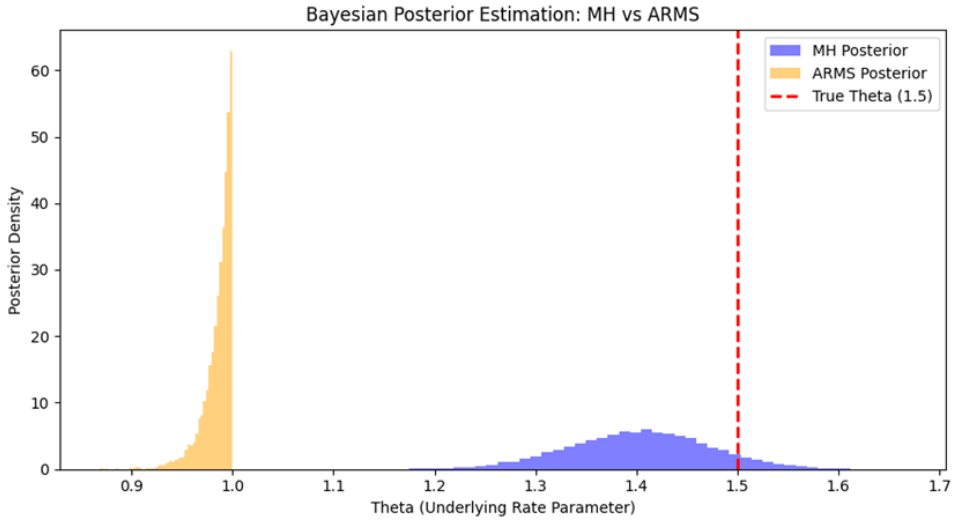


Figure 13: Bayesian Posterior Estimation - MH vs ARMS on Poisson log-rate model. Both posteriors concentrate around the true parameter $\theta = 1.5$ (red dashed), confirming correct sampling. ARMS achieves 100% acceptance rate; RWMH achieves 47.52%.

Table 4: Bayesian Posterior Estimation Performance

Algorithm	**Acceptance Rate**	**Time (s)**	**ESS (of 45,001)**	**ESS/Second**
RWMH	47.52%	0.5998	9,861.67	16,440.59
ARMS	100.00%	5.9777	9,844.21	1,646.82

Both algorithms produced identical ESS ($\sim 9,850$ effective samples), confirming they sampled from the same posterior. However, ARMS required $10\times$ more wall-clock time per iteration due to hull evaluation overhead, resulting in $10\times$ lower ESS/second (1,647 vs 16,441). The conclusion is that on log-concave posteriors where RWMH can be well tuned, ARMS's overhead does not pay off. ARMS's advantage is most pronounced on multimodal or non-log-concave targets where the fixed Gaussian proposal fails systematically.

5.10 Comprehensive Results Summary

Table 5: Comprehensive Results Summary

Experiment	Algorithm	Key Metric	Value
RWMH N(0,1)	RWMH	Acceptance rate	50.22%
ARMS bimodal test	ARMS	Support points (final)	∼10 (from 5)
SA non-convex landscape	RWMH+SA	Best state (true: 0)	0.001
SA non-convex landscape	ARMS+SA	Best state (true: 0)	-0.010
SA Rastrigin single run	RWMH+SA	Best state (true: 0)	-0.002
SA Rastrigin single run	ARMS+SA	Best state (true: 0)	-0.0004
SA Ackley single run	RWMH+SA	Best state (true: 0)	0.000124
SA Ackley single run	ARMS+SA	Best state (true: 0)	0.000002
Rastrigin 10-seed	RWMH	Mean ± std	-0.000121 ± 0.000146
Rastrigin 10-seed	ARMS (Exact)	Mean ± std	-0.056137 ± 0.153187
Ackley 10-seed	RWMH	Mean ± std	-2.073665 ± 5.768930
Ackley 10-seed	ARMS	Mean ± std	-0.000004 ± 0.000003
Gaussian Mixture 10-seed	RWMH	Mean ± std	-1.808778 ± 0.101222
Gaussian Mixture 10-seed	ARMS	Mean ± std	-1.753367 ± 0.000306
ARMS ablation 10k grid+jitter	ARMS	Median (true: 0)	-0.000000
Bayesian Poisson	RWMH	ESS/second	16,440.59
Bayesian Poisson	ARMS	ESS/second	1,646.82

6 Conclusions and Key Observations

1. **Both algorithms converge reliably when embedded in SA:** Both RWMH and ARMS located states within 0.01 log-density units of the theoretical global maximum on all benchmark functions, validating that high-temperature exploration successfully escapes local optima. This validates the theoretical claim that sufficiently high T_0 allows the SA exploration phase to overcome the basin of attraction of local optima.

2. **ARMS dominates on multimodal and non-log-concave targets:** On Ackley (mean -0.000004 vs -2.07) and Gaussian Mixture (std 0.000306 vs 0.101222), ARMS dramatically outperforms RWMH. The adaptive envelope correctly identifies target structure while the fixed Gaussian proposal fails systematically on these landscapes.

3. **RWMH is computationally more efficient on log-concave posteriors:** On the Bayesian Poisson problem, RWMH achieved 10× higher ESS/second (16,440 vs 1,647). Hull evaluation overhead does not pay off when the target is well-matched to a Gaussian proposal.

4. **SA convergence is not guaranteed under geometric cooling:** SA is a time-inhomogeneous Markov chain; standard MCMC ergodic guarantees do not apply. Geometric cooling success depends on T_0 being large enough relative to barrier depth. Logarithmic cooling provides a theoretical guarantee but at the impractical $\mathcal{O}(1/\ln n)$ convergence rate.

5. **Grid resolution matters for ARMS optimization quality:** The ablation study showed 10,000-point grid ARMS with jitter outperforms exact analytical ARMS due to superior continuous coverage. Continuous jitter enables exploration between discrete support points.

6. **ARMS is more robust to cooling rate variation:** ARMS achieved near-zero log-density at all tested η values; RWMH degraded at $\eta = 0.99$. The adaptive proposal reduces dependence on a single fixed bandwidth hyperparameter.

7 Overall Learning Outcomes

- **Ergodic Markov chains and detailed balance:** Implementing RWMH and ARMS from scratch required tracing the detailed balance condition $\pi(x)P(x, y) = \pi(y)P(y, x)$ through both the single-stage MH case and the more complex two-stage ARMS case, making the connection between detailed balance, stationarity, and ergodicity concrete.
- **Log-space numerical computation:** All acceptance probabilities are computed in log-space to avoid underflow when densities are very small. This practically essential technique was learned through direct experience debugging overflow errors in early prototype code.
- **Piecewise linear approximation and adaptive algorithms:** Building the ARMS hull required careful handling of sorted support point sets and bisection-based insertion. The visual comparison against the true log-density made the necessity of adaptive hull updates immediately apparent.
- **Time-inhomogeneous Markov chains in SA:** The Markov property is preserved (next state depends only on current state) but the transition kernel changes over time as T decreases. Understanding this distinction clarified why SA lacks standard MCMC convergence guarantees.
- **Convergence diagnostics:** Implementing Gelman-Rubin $\hat{R}$ from scratch, including within-chain and between-chain variance decomposition, and computing ESS via FFT-based autocorrelation, gave concrete quantitative understanding of MCMC convergence.
- **ESS and computational efficiency tradeoffs:** The Bayesian Poisson experiment revealed that algorithmic sophistication does not always translate to computational efficiency - ARMS's advantage is most valuable precisely where the fixed Gaussian proposal fails.

8 Limitations of the Proposed Study

- **Univariate setting:** All experiments are 1D. ARMS was originally designed for univariate targets within a Gibbs sampler. Results cannot be directly extrapolated to high-dimensional problems, where extension requires algorithmic variants such as Hit-and-Run ARMS.
- **Incomplete adaptation in standard ARMS:** Standard ARMS (Gilks et al., 1995) only updates support points on Stage 1 rejections. Martino et al. (2015) showed IA2RMS, which updates on both rejection types, provides lower autocorrelation and faster convergence.
- **No convergence guarantee under geometric cooling:** Due to inherent non-stationarity of SA, geometric cooling provides no theoretical guarantee. Some runs may terminate in a local optimum if T_0 is insufficient relative to barrier depth.
- **ARMS stochastic performance on Rastrigin:** The exact analytical ARMS occasionally traps (Run 2, seed 86837: final log-density -0.515), reflecting sensitivity to initialization in the early-epoch crude-envelope phase.

9 Future Directions and Scope for Further Investigations

- **IA2RMS implementation:** Implementing the Improved Adaptive Rejection Metropolis Sampling algorithm (Martino et al., 2015) would directly address the incomplete adaptation limitation by updating support points after both Stage 1 and Stage 2 rejections.
- **Extension to multivariate targets:** Implementing Hit-and-Run ARMS (HARARMS; Zhang, Chan, and Gao, 2015) would allow application to 2D and higher-dimensional versions of Rastrigin and Ackley, providing a direct scalability test.
- **Parallel tempering:** Running multiple chains at different temperatures simultaneously with swap moves between adjacent chains would make optimization more robust to T_0 choice and accelerate convergence on highly multimodal landscapes.
- **Adaptive cooling schedule:** Instead of a fixed η, adapting the cooling rate based on observed acceptance rates to maintain a target acceptance ratio could provide both practical efficiency and stronger convergence properties.
- **Real Bayesian inference application:** Extending to a genuine multiparameter Bayesian regression problem with a multimodal posterior would demonstrate practical relevance beyond synthetic benchmarks.

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